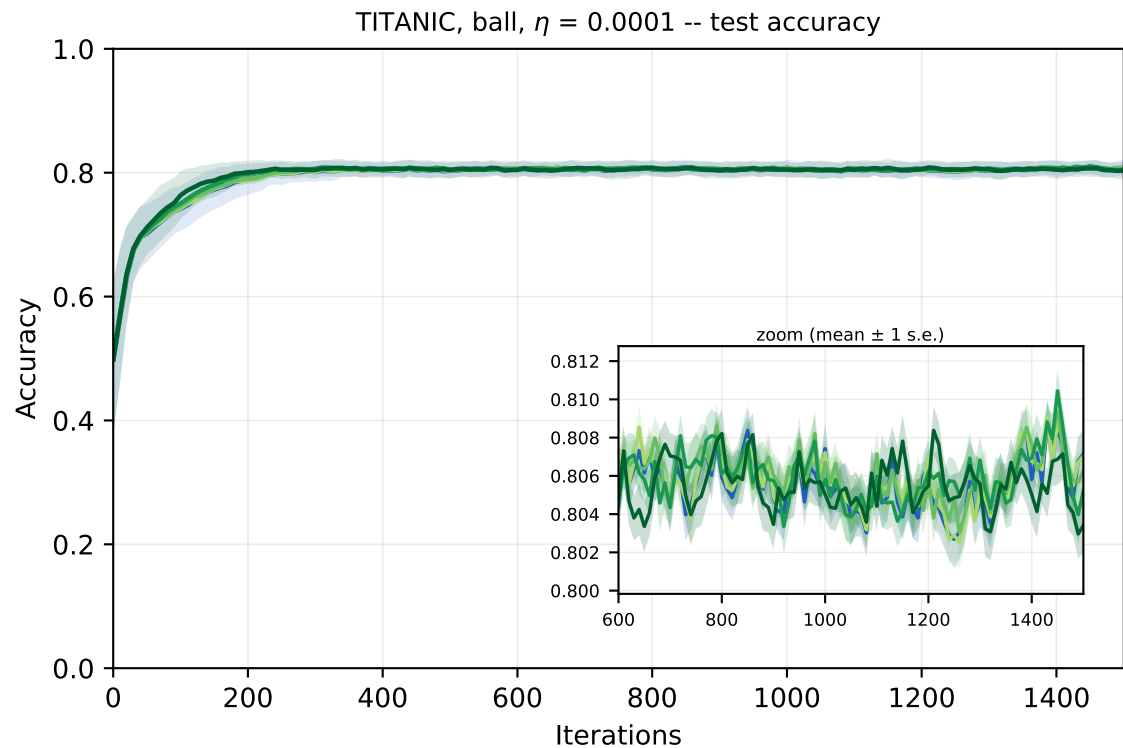
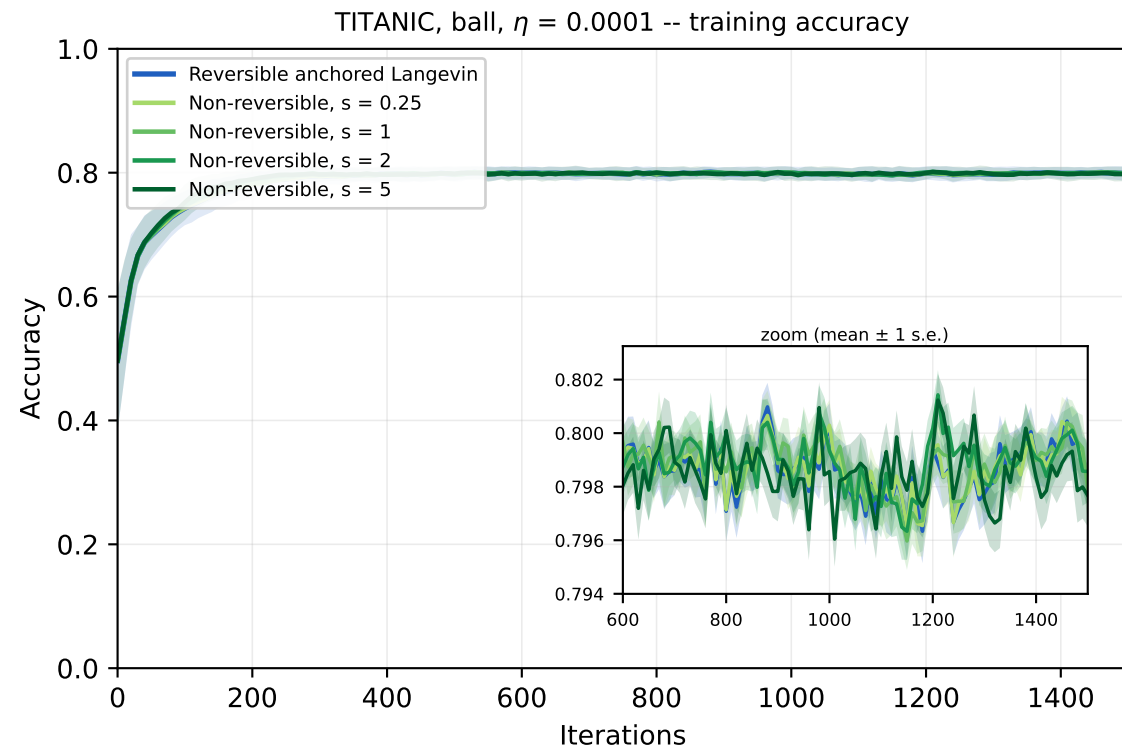


titanic_ball: accuracy vs block strength at $\eta = 0.0001$



TITANIC -- ball $\|w\|^2 \leq 2$ (radius $\sqrt{2}$). $d = 10$ (intercept w_0 + 9 features). $U = \sum \text{softplus} - y x \cdot w + w_0^2 / (2 \sigma^2) + \lambda \sum_{j \geq 1} |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \delta^2}$; $a = \exp(U - U_0)$. EXACT gradient. $\sigma = 10$, $\lambda_{\text{lasso}} = 7.12 = 0.01 n_{\text{train}}$, $\delta = 0.0108$ (a in $[0.50, 1]$). Step $\eta = 0.0001$. $R = 100$ independent replicates on one fixed stratified 80/20 split ($n_{\text{train}} = 712$, $n_{\text{test}} = 179$). Bands: mean $\pm$ 1 SD ($\text{ddof} = 1$) for the reversible arm and for $s = 5$ -- repeat-run variability at a fixed split, not a confidence or credible interval. The whole $\eta \times s$ grid is reported in results/grid/grid.csv; nothing here was selected.